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Exhibit R-2, RDT&E Budget Item Justification: PB 2026 Defense Advanced Research Projects Agency	Date: June 2025
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Appropriation/Budget Activity	R-1 Program Element (Number/Name)											
0400: <i>Research, Development, Test & Evaluation, Defense-Wide / BA 1: Basic Research</i>	PE 0601101E / <i>DEFENSE RESEARCH SCIENCES</i>											
COST (\$ in Millions)	Prior Years	FY 2024	FY 2025	FY 2026 Base	FY 2026 OOC	FY 2026 Total	FY 2027	FY 2028	FY 2029	FY 2030	Cost To Complete	Total Cost
Total Program Element	0.000	280.494	293.145	0.000	-	0.000	-	-	-	-	-	-
CCS-02: <i>MATH AND COMPUTER SCIENCES</i>	0.000	160.158	188.187	0.000	-	0.000	-	-	-	-	-	-
ES-01: <i>ELECTRONIC SCIENCES</i>	0.000	5.802	4.768	0.000	-	0.000	-	-	-	-	-	-
ES-02: <i>BEYOND SCALING SCIENCES</i>	0.000	43.607	44.665	0.000	-	0.000	-	-	-	-	-	-
MS-01: <i>MATERIALS SCIENCES</i>	0.000	66.406	55.525	0.000	-	0.000	-	-	-	-	-	-
TRS-01: <i>TRANSFORMATIVE SCIENCES</i>	0.000	4.521	0.000	0.000	-	0.000	-	-	-	-	-	-

A. Mission Description and Budget Item Justification

The efforts described in this Program Element (PE) address the Basic Research associated with the Defense Research Sciences Program that provides the technical foundation for long-term National Security enhancement through the discovery of new phenomena and the exploration of the potential of such phenomena for Defense applications. This PE supports the scientific study and experimentation that is the basis for more advanced knowledge and understanding in information, electronic, mathematical, computer, and materials sciences. This PE also supports innovation and robust transition planning in the technology cycle by working with entrepreneurs to increase the likelihood that DARPA funded technologies take root in the U.S. and provide new capabilities for national defense.

The Math and Computer Sciences project supports scientific study and experimentation on new mathematical and computational algorithms, models, and mechanisms in support of long-term national security objectives. Modern analytic and information technologies enable important new military capabilities and drive the productivity gains essential to U.S. economic competitiveness. Conversely, new classes of threats, in particular threats that operate in or through the cyber and information domain, put military systems, critical infrastructure, and the civilian economy at risk. This project aims to magnify these opportunities and mitigate these threats by leveraging emerging mathematical and computational capabilities including artificial intelligence (AI), computational social science, machine learning and reasoning, data science, quantum science, complex systems modeling and simulation, and theories of computation and programming. The basic research conducted under the Math and Computer Sciences project will produce breakthroughs that enable new capabilities for national and homeland security.

The Electronic Sciences project is for basic exploration of electronic and optoelectronic devices, circuits, and processing concepts to meet the military's need for near real-time information gathering, transmission, and processing. In seeking to continue the phenomenal advancement in microelectronics innovation that has characterized the last few decades, the project will provide DoD with new, improved, or potentially revolutionary device options for accomplishing these critical functions. The resulting technologies will help maintain knowledge of the enemy, communicate decisions based on that knowledge, and substantially improve the cost and performance of military systems. Research areas include analog, mixed signal, and photonic circuitry for communications and other applications; alternative computer architectures;

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and magnetic components to reduce the size of Electromagnetic (EM) and sensing systems. Other research could support field-portable electronics with reduced power requirements, and new approaches to nanometer-scale structures, molecules, and devices.						
The Beyond Scaling Sciences project supports investigations into materials, devices, and architectures to provide disruptive improvements in electronics performance that can be realized by techniques other than transistor scaling. Examples include circuit specialization, non-volatile memory devices that combine computation and memory, and new automated design tools using machine learning. Additionally, new design and manufacturing advances for three-dimensional microelectronics integration will underpin continued performance improvements as silicon transistor scaling plateaus.						
The Materials Sciences project provides the fundamental research that underpins the design, development, assembly, and optimization of advanced materials, devices, and systems for DoD applications in areas such as robust diagnostics and therapeutics, novel energetic materials, and complex hybrid systems.						
The Transformative Sciences project supports research and analysis that leverages converging technological forces and transformational trends in information-intensive subareas of life sciences, data sciences, and manufacturing. Innovative technologies developed in this project will address multiple DoD challenges such as identification of and adaptation to emerging threats, access to DoD relevant critical materials for manufacturing and warfighter readiness. Successful programs in this project will integrate diverse disciplines and engineer complex biological systems to detect novel threat agents, accelerate warfighter injury recover, accelerate recovery of DoD natural resources following natural disaster, and develop new platform materials and manufacturing processes.						
Beginning in FY 2026, efforts in this PE will be funded in PE 0601122E, Emerging Opportunities.						
B. Program Change Summary (\$ in Millions)		FY 2024	FY 2025	FY 2026 Base	FY 2026 OOC	FY 2026 Total
Previous President's Budget		311.531	303.830	332.425	-	332.425
Current President's Budget		280.494	293.145	0.000	-	0.000
Total Adjustments		-31.037	-10.685	-332.425	-	-332.425
• Congressional General Reductions		-7.500	-10.685			
• Congressional Directed Reductions		0.000	0.000			
• Congressional Rescissions		0.000	0.000			
• Congressional Adds		0.000	0.000			
• Congressional Directed Transfers		0.000	0.000			
• Reprogrammings		-13.722	0.000			
• SBIR/STTR Transfer		-9.815	0.000			
• TotalOtherAdjustments		-	-	-332.425	-	-332.425
Change Summary Explanation						
FY 2024: Decrease reflects Congressional general reduction to reduce carryover in Next Generation Microelectronics - Advanced Manufacturing Science, SBIR/STTR transfer and reprogrammings.						
FY 2025: Decrease reflects Congressional general reduction for an unjustified request.						